Data and Systems Engineering

	Normative 4-year Degree
Major	Data and Systems Engineering 數據與系統工程
Degree / Award Title	Bachelor of Science in Data and Systems Engineering 理學士(數據與系統工程)
Offering Academic Unit	Department of Data Science
Normal Period of Study	4 years
Maximum Period of Study	8 years
Credit Units Required for Graduation	121 credit units

Note: The following curriculum information is subject to periodic review and changes.

GE Requirements (31 credit units) Catalogue Term : Semester A 2025/26

		Credit Units
GE University Requirements	GE1401 University English	3
	GE2410 English for Engineering	3
	GE1501 Chinese Civilisation - History and Philosophy	3
	GE1601 Whole-Person Development	1
GE Distributional Requirements	Take at least 1 course from each of the three distributional areas: Area 1: Arts and Humanities Area 2: Study of Societies, Social and Business Organisations Area 3: Science and Technology Visit the GE website for more course information.	12

College-specified Courses	CS1315 Introduction to Computer Programming	3
	SDSC2003 Human Contexts and Ethics in Data Science	3
	CS3402 Database Systems	3

College / School Requirements (8 credit units) **Catalogue Term : Semester A 2025/26**

Course Code	Course Title	Credit Units
MA1503	Linear Algebra with Applications	4
MA1508	Calculus	4

Major Requirements (73 credit units) **Catalogue Term : Semester A 2026/27**

Notes:

The catalogue term of the major requirement that students will follow will be the effective term of the declared/allocated major.

For normative 4-year degree students who will join the majors allocation exercise, the catalogue term of major requirement will be one year after admission.

For advanced standing students and 4-year degree students who already have a major at the time of admission, the catalogue term of major requirement will be the same as their admission term.

1. Core Courses

(49 credit units for students with no stream)
(58 credit units for ACT stream)

Course Code	Course Title	Credit Units	Remarks
PHY1201	General Physics I	3	
SDSC1001	Introduction to Data Science	2	
GE2339	Smart City - a Systems Engineering Perspective	3	
MA2508	Multi-variable Calculus	4	
MA2510	Probability and Statistics	3	
SDSC2001	Python for Data Science	4	
SDSC2002	Convex Optimization	3	
SDSC2004	Data Visualization	3	
SDSC2102	Statistical Methods and Data Analysis	3	
SDSC3002	Data Mining	3	
SDSC3006	Fundamentals of Machine Learning I	3	
SDSC3008	Systems Dynamics and Control	3	
SDSC3025	Internship for Flagship Programme	3	ACT stream only

SDSC3026	International Professional Development	3	ACT stream only
SDSC3060	Operations Research	3	
SDSC4016	Fundamentals of Machine Learning II	3	ACT stream only
CS4480	Data-Intensive Computing	3	
SDSC4116	Data Science Capstone	6	

1. Electives

(24 credit units for students with no stream)
(15 credit units for ACT stream)

Course Code	Course Title	Credit Units	Remarks
SDSC0001	Internship	0	General
SDSC0002	Internship	0	General
SDSC0003	Internship	0	General
SDSC0004	Internship	0	General
CS2334	Data Structures for Data Science	3	General
CS3201	Computer Networks	3	Industrial Artificial Intelligence
CS3273	Data Protection and System Security	3	Industrial Artificial Intelligence
CS3342	Software Design	3	General
CS3383	AI for Software Engineering	3	General
LT3233	Computational Linguistics	3	General
SDSC3001	Big Data: The Arts and Science of Scaling	3	Industrial Artificial Intelligence
SDSC3003	Blockchain	3	FinTech & Decision Analytics
SDSC3004	Computational Optimization	3	Industrial Artificial Intelligence
SDSC3013	Introduction to Social Media Analytics	3	General
SDSC3014	Introduction to Sharing Economy	3	Smart City
SDSC3018	Introduction to Internet of Things	3	Smart City
SDSC3019	Introduction to Networked Life and Data Science	3	Smart City
SDSC3020	Engineering Economics	3	FinTech & Decision Analytics
SDSC3022	Financial Data Analytics for Investments	3	FinTech & Decision Analytics
SDSC3023	Data Science Applications in Portfolio Risk Analysis	3	FinTech & Decision Analytics
SDSC3027	Smart Logistics and Transportation	3	Smart City
SDSC3080	Internship	3	General, Not available for ACT stream

SDSC3102	Quality Technologies	3	General
SDSC3105	Bayesian Analysis	3	General
CS4286	Internet Security and e-Commerce Protocols	3	FinTech & Decision Analytics
CS4486	Artificial Intelligence	3	Industrial Artificial Intelligence
SDSC4001	Foundation of Reinforcement Learning	3	General
SDSC4009	Data Intelligence in Action	3	Industrial Artificial Intelligence
SDSC4016	Fundamentals of Machine Learning II	3	Industrial Artificial Intelligence, Core course for ACT stream
SDSC4018	AI in Systematic Trading	3	FinTech & Decision Analytics
SDSC4021	Advanced Internet of Things	3	Industrial Artificial Intelligence
SDSC4024	Project Management and Analysis	3	FinTech & Decision Analytics
SDSC4026	Systems Modelling and Simulation	3	Industrial Artificial Intelligence
SDSC4051	Facilities and Distribution Management	3	Smart City
SDSC4064	Reliability Engineering	3	General
SDSC4070	Large Language Models	3	General
SDSC4103	Decision Analytics and Risk Management	3	FinTech & Decision Analytics
SDSC4107	Financial Engineering and Analytics	3	FinTech & Decision Analytics
SDSC4109	Smart Manufacturing and Automation	3	Smart City
SDSC4110	Statistical Design and Analysis of Experiments	3	General